

A no-go theorem and a bounded exclusion of exceptional points in the massive scalar spectrum of doubly rotating five-dimensional Myers–Perry black holes

AUTHOR NAME NOT ASSIGNED^{1,*}

¹AFFILIATION NOT ASSIGNED. ORCID NOT ASSIGNED.

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We ask whether the enhanced symmetry of the equal-spin surface organizes defective (exceptional) degeneracies in the massive scalar quasinormal spectrum of the asymptotically flat, doubly rotating five-dimensional Myers–Perry black hole. Our first result is exact. Because ∂_ϕ and ∂_ψ remain Killing for arbitrary rotation parameters, the quasinormal pencil admits an invariant decomposition into azimuthal sectors (m_1, m_2) , and the Jordan chains of such a decomposition are the union of the chains of its summands. A coincidence of frequencies drawn from distinct sectors therefore cannot by itself create defectiveness: any Jordan chain of length greater than one must already occur within a single fixed sector. Through the Keldysh expansion of the resolvent this statement is equivalent to the assertion that such a coincidence cannot raise the order of the ringdown Green-function pole above one, so it has direct observational content. The equal-spin multiplets, whose members carry distinct (m_1, m_2) , are accordingly protected from *hybridization*: their frequencies may and do separate as parameters vary, but they cannot coalesce into a shared defective block. A residual Klein four-group acting on parameters *and* sector labels renders diagonal sectors $m_1 = m_2$ exactly even in the spin asymmetry δ ; when the associated discriminant curve crosses $\delta^2 = 0$ transversally, the physical exceptional branches must occur in mirror pairs and terminate quadratically on the equal-spin surface. Our second result is numerical and explicitly bounded. Searching the one channel the theorem leaves open — collisions between distinct (ℓ, N) branches inside a fixed sector — over an atlas of 50 558 continued modes in seven sectors, we find no exceptional point within the declared domain, where the minimum separation between tracked branches is 0.5116 in units of the inverse horizon scale and the eigenvalue condition number remains finite at 8.73×10^3 . The ten tightest candidates are classified as avoided crossings by argument-principle, monodromy and Puiseux tests. The searched domain lies entirely outside the superradiant regime, every tracked mode exceeding the critical frequency $m_1\Omega_a + m_2\Omega_b$ by at least 1.07. We emphasize that this is an exclusion within a declared domain: higher overtones, higher angular branches, the superradiant regime, and the quasisresonant regime — where no recurrence-free solver is applicable — were not searched.

I. INTRODUCTION

A. Defective degeneracies and their observational content

Quasinormal modes are the resonances of a black hole: the poles of the retarded Green function of the perturbation operator, and the observable content of the ringdown [1–4]. Because the governing operator is not self-adjoint — energy leaks through the horizon and out to infinity — its spectral theory admits structures with no Hermitian counterpart. The sharpest of these is the *exceptional point* (EP): a value of the external parameters at which two or more eigenvalues *together with their eigenvectors* coalesce, so that the generator becomes defective and cannot be diagonalized [5–7].

The physical consequences are not decorative. If the perturbation problem is written as a holomorphic operator pencil $T(\omega)$, the Keldysh expansion of the resolvent about an eigenvalue ω_0 carrying a Jordan chain of length q produces a pole of order exactly q (Appendix C). Inverse Laplace transformation then yields a time-domain

response

$$\psi(t) \sim \left(A_0 + A_1 t + \cdots + A_{q-1} t^{q-1} \right) e^{-i\omega_0 t}, \quad (1)$$

so that at an EP2 the ringdown acquires a secular $t e^{-i\omega_0 t}$ term [8, 9], the quasinormal expansion ceases to be a basis, and the excitation factors of the participating modes diverge individually while their sum stays finite. In the neighbourhood of an EP2 the eigenvalues respond to a perturbation of size ϵ as $\mathcal{O}(\epsilon^{1/2})$ rather than $\mathcal{O}(\epsilon)$, which is the same square-root sensitivity that drives the pseudospectral instability of black-hole spectra [10–12]. For black-hole spectroscopy this matters directly: overtone-resolved ringdown fits presuppose a discrete, diagonalizable spectrum, and the presence of a nearby exceptional structure would invalidate that presupposition rather than merely perturb it.

Such structures are by now documented across black-hole perturbation theory. Branching of quasinormal modes was identified for nearly extremal Kerr [13]; eigenvalue repulsions organize the Kerr–Newman spectrum and connect the Kerr and Reissner–Nordström families [14–16]; genuine exceptional points, hysteresis, and ergodic spectral flow appear for massive scalars on Kerr [17–19]; and the resonant excitation associated with near-degenerate pairs has been analysed in detail [20–24]. Most directly relevant here, continuous *exceptional*

* corresponding.author@NOT-ASSIGNED

lines have been established for massive scalars on Kerr–de Sitter and Myers–Perry–de Sitter, where the near-Nariai limit reduces the radial problem to Pöschl–Teller form and permits a semi-analytic treatment [25–27].

The non-Hermitian literature supplies the counting that organizes the search. An EP2 is the vanishing of a complex discriminant, hence two real conditions: codimension two. A one-parameter family therefore meets exceptional points only accidentally, a two-parameter family meets them at isolated points, and a three-parameter family meets them along curves [6]. Symmetry modifies this counting: parity–time or generalized chiral symmetry reduces the codimension of *higher-order* points, making third-order points generic in two parameters rather than the four otherwise required [28], and symmetry can protect entire exceptional chains in lattice models [29].

B. Why five-dimensional Myers–Perry

The five-dimensional Myers–Perry black hole [30, 31] is an unusually favourable laboratory for asking whether that machinery has a gravitational realization. Four features make it so.

First, the massive Klein–Gordon equation separates exactly for arbitrary a and b [32, 33], a property extending to Dirac and massive vector fields [33, 34], so the spectral problem reduces to a coupled pair of ordinary differential equations rather than a genuinely two-dimensional eigenvalue problem.

Second, the massive scalar spectrum with two arbitrary spins has been computed [35–37], including the long-lived branches that appear when μ becomes comparable to the inverse horizon scale, so a numerical baseline exists.

Third, the parameter space has *exactly* the dimension at which codimension-two sets become generic. The background supplies two independent rotation parameters and the field supplies a mass, giving the three real parameters (s, δ, μ) defined below. Four-dimensional Kerr, by contrast, offers only (a, μ) , so exceptional points there are isolated rather than curves. If symmetry-organized exceptional structure exists anywhere in asymptotically flat black-hole perturbation theory, this is where the counting says it should be easiest to find.

Fourth, and decisively for the question we ask, the isometry group is enhanced when the two angular momenta coincide,

$$\mathbb{R} \times U(1) \times U(1) \longrightarrow \mathbb{R} \times SU(2) \times U(1), \quad (2)$$

and the metric becomes cohomogeneity-one [38]. Equation (2) produces exact degeneracies in the perturbation spectrum, and the equal-spin surface is a symmetric locus embedded in the three-dimensional parameter space — precisely the configuration in which one expects symmetry-protected exceptional structure to appear.

Higher-dimensional rotating black holes are of interest in their own right, as the arena in which black-hole

uniqueness fails, as the setting of the Gregory–Laflamme and ultraspinning instabilities [39, 40], and through the large- D expansion. Our motivation here is more specific: we use the system as a controlled setting in which the parameter counting is favourable and an exact symmetry statement is available, and we regard the methodological conclusions of Secs. VI and IX as transferable to the four-dimensional problem.

C. Statement of results

The natural expectation, and the standard mechanism in non-Hermitian systems [28, 29], is that the degeneracies created by (2) unfold into exceptional points once the symmetry is broken by $\delta \neq 0$. We show that this expectation fails, and that it fails for an exact reason.

We claim the following. (i) An exact structural theorem (Theorem 5): a coincidence of frequencies from distinct azimuthal sectors cannot by itself produce a Jordan chain, hence cannot raise the order of the ringdown Green-function pole. (ii) An exact symmetry classification (Table II), with a conditional termination law for diagonal-sector exceptional loci (Proposition 8). (iii) A *bounded* numerical exclusion of exceptional points within an explicitly declared parameter domain, Eq. (54). Claim (iii) is not a statement about the system as a whole: overtones $N > 3$, angular branches beyond those listed, the region $s > 0.45$ or $\mu > 1.8$, the superradiant regime, and the quasisonant regime lie outside it, and we make no claim there.

The paper is organized as follows. Section II fixes the geometry and its thermodynamics. Section III derives the separated system, its singular structure, and the flux identity that defines the superradiant regime. Section IV proves the sector decomposition and the no-go theorem. Section V classifies the residual symmetry. Section VI develops the local spectral theory and establishes which diagnostics are admissible — a question on which we found two natural choices to be inadmissible. Section VII describes the numerical formulations, Sec. VIII the atlas, and Sec. IX the exclusion. Sections X–XI treat the quasisonant limit and the observational implications, and Sec. XII the limitations.

II. BACKGROUND GEOMETRY AND HORIZON THERMODYNAMICS

A. Metric and horizons

In Boyer–Lindquist-like coordinates $(t, r, \theta, \phi, \psi)$ with $\theta \in [0, \pi/2]$ and $\phi, \psi \in [0, 2\pi)$, signature $(- + + +)$, the

Myers–Perry metric with two rotation parameters reads

$$ds^2 = -dt^2 + \frac{M}{\Sigma} \left(dt - a \sin^2 \theta d\phi - b \cos^2 \theta d\psi \right)^2 + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 + (r^2 + a^2) \sin^2 \theta d\phi^2 + (r^2 + b^2) \cos^2 \theta d\psi^2, \quad (3)$$

with

$$\Sigma = r^2 + a^2 \cos^2 \theta + b^2 \sin^2 \theta, \quad \Pi = (r^2 + a^2)(r^2 + b^2), \quad (4)$$

and $\Delta = (\Pi - Mr^2)/r^2$. Here M is the Myers–Perry mass parameter, of dimension (length)², *not* the scalar mass μ ; we set $M = 1$, which fixes the length unit, and we verify $\sqrt{|g|} = r \Sigma \sin \theta \cos \theta$ as a check on the symbolic derivation. The ADM mass and angular momenta are $M_{\text{ADM}} = 3\pi M/8$, $J_\phi = \frac{2}{3} M_{\text{ADM}} a$, $J_\psi = \frac{2}{3} M_{\text{ADM}} b$.

With $z = r^2$ the horizon polynomial is

$$P(z) = (z + a^2)(z + b^2) - Mz = (z - z_+)(z - z_-), \quad (5)$$

whence

$$z_\pm = \frac{1}{2} \left[(M - a^2 - b^2) \pm \sqrt{(M - a^2 - b^2)^2 - 4a^2 b^2} \right], \quad (6)$$

with $z_+ z_- = a^2 b^2$ and $z_+ + z_- = M - a^2 - b^2$. A horizon exists iff $M \geq (|a| + |b|)^2$, and we use

$$\mathcal{E} \equiv M - (|a| + |b|)^2 \quad (7)$$

as the extremality measure, $\mathcal{E} = 0$ at extremality. Note that (6) implies $z_- \leq M/4$ for any sub-extremal configuration, so the inner horizon radius satisfies $r_- \leq \frac{1}{2} \sqrt{M}$; we quote r_- rather than z_- throughout, since a domain restriction on one is a different restriction on the other.

Throughout we use the adapted rotation coordinates

$$s = \frac{a+b}{2}, \quad \delta = \frac{a-b}{2}, \quad (8)$$

so $\delta = 0$ is the equal-spin surface on which (2) holds and $s = 0$ is the counter-rotating surface $a = -b$.

B. Horizon thermodynamics

The horizon generators are $\xi = \partial_t + \Omega_a \partial_\phi + \Omega_b \partial_\psi$ with

$$\Omega_a = \frac{a}{z_+ + a^2}, \quad \Omega_b = \frac{b}{z_+ + b^2}, \quad (9)$$

and the surface gravity and Hawking temperature are

$$\kappa = \frac{r_+(z_+ - z_-)}{(z_+ + a^2)(z_+ + b^2)}, \quad T_H = \frac{\kappa}{2\pi}. \quad (10)$$

The horizon is a squashed three-sphere of area

$$\mathcal{A} = \frac{2\pi^2}{r_+} (z_+ + a^2)(z_+ + b^2), \quad S_{\text{BH}} = \frac{\mathcal{A}}{4}, \quad (11)$$

s	δ	r_+	T_H	Ω_a	Ω_b	S_{BH}
0.00	0.00	1.0000	0.15915	0.0000	0.0000	4.9348
0.20	0.00	0.9583	0.15222	0.2087	0.2087	4.7288
0.42	0.00	0.7713	0.11196	0.5445	0.5445	3.8062
0.34	0.20	0.8249	0.12966	0.5556	0.2000	4.0705
0.42	0.20	0.7296	0.10849	0.6764	0.3789	3.6002
0.42	-0.20	0.7296	0.10849	0.3789	0.6764	3.6002

TABLE I. Horizon data at the corners of the searched domain, in units $M = 1$. The last two rows are exchanged by the generator E of Sec. V, as they must be.

which reduces to $\mathcal{A} = 2\pi^2 r_+^3$ and $T_H = 1/(2\pi r_+)$ in the Schwarzschild–Tangherlini limit $a = b = 0$; both limits are verified numerically. Table I records these quantities at the corners of the searched domain. The temperature varies by only 30% across the domain while the horizon angular velocity ranges from zero to 0.68, so the domain is a rotation-dominated rather than temperature-dominated slice of parameter space.

III. THE SEPARATED SYSTEM

A. Separation

With the mode ansatz

$$\Phi = e^{-i\omega t + im_1 \phi + im_2 \psi} R(r) S(\theta), \quad m_1, m_2 \in \mathbb{Z}, \quad (12)$$

the equation $(\square - \mu^2)\Phi = 0$ separates exactly. Since $\Phi \propto e^{-i\omega t}$, damped modes have $\text{Im } \omega < 0$. The angular equation is

$$\frac{1}{\sin \theta \cos \theta} \frac{d}{d\theta} \left[\sin \theta \cos \theta \frac{dS}{d\theta} \right] + \left[\Lambda + (\omega^2 - \mu^2)(a^2 \cos^2 \theta + b^2 \sin^2 \theta) - \frac{m_1^2}{\sin^2 \theta} - \frac{m_2^2}{\cos^2 \theta} \right] S = 0, \quad (13)$$

and the radial equation

$$\frac{1}{r} \frac{d}{dr} \left[r \Delta \frac{dR}{dr} \right] + \left[\frac{W^2}{r^4 \Delta} - \frac{G^2}{r^2} - (a^2 + b^2) \omega^2 + 2\omega(am_1 + bm_2) - \mu^2 r^2 - \Lambda \right] R = 0, \quad (14)$$

with

$$W = \Pi \omega - m_1 a(r^2 + b^2) - m_2 b(r^2 + a^2), \quad G = ab\omega - am_2 - bm_1. \quad (15)$$

The separation constant is normalized by $\Lambda = \ell(\ell + 2)$ at $a = b = 0$. We re-derived (13)–(15) symbolically from (3) in exact rational arithmetic rather than transcribing them, and the derivation is re-executed as an automated test; the route is given in Appendix A.

B. The angular problem and two exactly degenerate surfaces

Substituting $u = \cos^2\theta$ in (13) converts it to a Jacobi-type equation on $u \in [0, 1]$ with regular singular points at $u = 0, 1$ and an irregular point at infinity. Writing

$$\hat{A} \equiv \Lambda + (\omega^2 - \mu^2)b^2, \quad c^2 \equiv (\omega^2 - \mu^2)(a^2 - b^2), \quad (16)$$

the problem becomes a five-dimensional spheroidal problem of the type classified in Ref. [41], with c^2 the spheroidicity and $\hat{A}$ the eigenvalue. Because $a^2 - b^2 = (a+b)(a-b) = 4s\delta$, one has the exact identity

$$\boxed{c^2 = 4s\delta(\omega^2 - \mu^2)} \quad (17)$$

which is not perturbative. Three consequences follow immediately.

(i) On $\delta = 0$ the spheroidicity vanishes identically for every s, μ and ω . The angular operator is then the round Laplacian on S^3 , with

$$\hat{A} = \ell(\ell + 2), \quad \ell = 2n + |m_1| + |m_2|, \quad n \geq 0, \quad (18)$$

and the full $(\ell + 1)^2$ -fold $SO(4)$ degeneracy.

(ii) The same holds on $s = 0$, the counter-rotating configuration $a = -b$. This is a *second* exactly degenerate surface, unrelated to (2), and the two intersect transversally at the origin of the (s, δ) plane.

(iii) The angular degeneracy is lifted only by the *product* $s\delta$, so the leading splitting of an $SO(4)$ multiplet is $\mathcal{O}(s\delta)$ rather than $\mathcal{O}(\delta)$.

Moreover, at $\delta = 0$ the radial equation (14) and the horizon condition depend on (m_1, m_2) only through

$$m = m_1 + m_2, \quad (19)$$

which is the algebraic origin of the multiplet structure of Sec. IV C.

C. Singular structure of the radial equation

In the variable $z = r^2$, Eq. (14) has apparent singularities at $z = 0$ and $z = z_{\pm}$ and an irregular singular point at infinity. The point $z = 0$ is only apparent: using $W(0) = abG$ and $P(0) = a^2b^2$, the would-be $1/z$ terms cancel identically whenever $ab \neq 0$, so $z = 0$ is an *ordinary* point. The equation is therefore of confluent Heun type, with exactly two regular singular points and one irregular point of rank one at infinity.

At $z = z_+$ the indicial equation is

$$\varrho^2 + \sigma_+^2 = 0, \quad \varrho = \pm i\sigma_+, \quad (20)$$

with

$$\sigma_+ = \frac{\omega - m_1\Omega_a - m_2\Omega_b}{2\kappa} \equiv \frac{\omega - \omega_c}{2\kappa}, \quad (21)$$

where the second equality defines the *critical frequency*

$$\omega_c \equiv m_1\Omega_a + m_2\Omega_b. \quad (22)$$

Equation (21) follows from the factorization $W(z_+) = (z_+ + a^2)(z_+ + b^2)(\omega - \omega_c)$, itself a consequence of (15) and (9). The quasinormal condition selects the ingoing branch $\varrho = -i\sigma_+$.

At large r , Eq. (14) reduces to $R'' + \frac{3}{r}R' + (\Omega^2 + \mathcal{O}(r^{-2}))R = 0$ with

$$\Omega \equiv \sqrt{\omega^2 - \mu^2}, \quad (23)$$

so that

$$R \sim e^{\pm i\Omega r} r^{-3/2} (1 + \mathcal{O}(r^{-1})). \quad (24)$$

The exponent $-3/2$ is exact and there is *no* Coulomb phase: because $D - 3 = 2$ the radial potential is an even function of r and possesses no $1/r$ tail. This is a structural difference from four-dimensional massive Kerr, where a logarithmic phase $\propto \mu^2 M \log r$ appears [42, 43], and it simplifies both the asymptotic matching and the continued-fraction construction of Sec. VII. Details are in Appendix B.

D. Flux identity and the superradiant regime

Equation (14) can be written in formally self-adjoint form $(r\Delta R')' + rV(r)R = 0$. For *real* ω and μ the potential V is real, so the Wronskian-like quantity

$$\mathcal{W}[R] = r\Delta(\bar{R}R' - R\bar{R}') \quad (25)$$

is independent of r . Evaluating (25) on a scattering solution that is ingoing at the horizon and a superposition of incoming and outgoing waves at infinity gives the standard flux balance

$$|\mathcal{R}|^2 = 1 - \frac{\sigma_+}{\Omega} \frac{(z_+ + a^2)(z_+ + b^2)}{\mathcal{N}}, \quad (26)$$

with $\mathcal{N} > 0$ a positive normalization depending only on the asymptotic amplitudes. The sign of the transmitted flux is therefore the sign of σ_+ , i.e. of $\omega - \omega_c$: for

$$0 < \omega < \omega_c = m_1\Omega_a + m_2\Omega_b \quad (27)$$

the reflected flux exceeds the incident flux and the mode extracts rotational energy from the black hole. This is the five-dimensional, two-spin analogue of Zel'dovich–Misner superradiance, and for a massive field it is the mechanism underlying superradiant bound-state instabilities [27, 36].

Two remarks matter for what follows. First, the entire searched domain lies *outside* (27): across the eleven branches tracked in Sec. X the smallest value of $\text{Re } \omega - \omega_c$ is $+1.07$, attained in sector (1, 1) where $\omega_c \leq 0.51$ while $\text{Re } \omega \geq 1.51$. Every mode we compute is consequently damped, as observed, and our exclusion says nothing about the superradiant sector. Second, σ_+ changes sign across (27), so the horizon boundary condition (21) changes character there; a search in that regime would require the boundary condition to be re-derived rather than continued.

E. The quasinormal problem as an operator pencil

Equations (13)–(14) with the boundary conditions (21) and (24) define a *nonlinear* eigenvalue problem: ω enters the coefficients, and Λ is itself a function of ω . We write it abstractly as a holomorphic operator pencil

$$T(\omega)u = 0, \quad (28)$$

and record the coupled scalar form used numerically,

$$F(\omega; s, \delta, \mu, m_1, m_2, \ell) = 0, \quad \Lambda = \Lambda(\omega; \dots), \quad (29)$$

in which Λ is recomputed at every evaluation. Freezing Λ while differentiating with respect to ω discards $\partial_\Lambda F \cdot d\Lambda/d\omega$ and is a standard route to spurious exceptional points; all derivatives here are total.

IV. SECTOR DECOMPOSITION AND THE NO-GO THEOREM

A. The torus action

No component of (3) depends on ϕ or ψ , so ∂_ϕ and ∂_ψ are Killing for *arbitrary* a, b — not merely on the equal-spin surface — and the torus $T^2 = S_\phi^1 \times S_\psi^1$ acts by

$$(\mathcal{R}_{\alpha\beta}u)(t, r, \theta, \phi, \psi) = u(t, r, \theta, \phi - \alpha, \psi - \beta). \quad (30)$$

Quasinormal eigenfunctions are not square integrable — by (24) they diverge at infinity whenever $\text{Im } \omega < 0$ — so a Fourier decomposition justified in $L^2(T^2)$ is not available. We therefore avoid L^2 entirely and work from the group action, stating the hypotheses explicitly, since they are what the argument consumes.

Proposition 1 (Invariant sector decomposition). *Let X be a Banach space carrying the pencil $T(\omega)$, and suppose*

(H1) *$\{\mathcal{R}_{\alpha\beta}\}$ is a uniformly bounded, strongly continuous representation of T^2 on X ;*

(H2) *$T(\omega)$ and all its ω -derivatives commute with every $\mathcal{R}_{\alpha\beta}$, and the domain of $T(\omega)$ is $\mathcal{R}$ -invariant.*

Then the isotypic projections

$$\Pi_{m_1 m_2} = \frac{1}{(2\pi)^2} \iint e^{-i(m_1 \alpha + m_2 \beta)} \mathcal{R}_{\alpha\beta} d\alpha d\beta \quad (31)$$

are bounded, mutually orthogonal idempotents commuting with $T(\omega)$, and they separate points:

$$\Pi_{m_1 m_2} u = 0 \quad \forall (m_1, m_2) \implies u = 0. \quad (32)$$

Proof. The integral (31) converges as a Bochner integral by (H1), with $\|\Pi_{m_1 m_2}\| \leq \sup_{\alpha\beta} \|\mathcal{R}_{\alpha\beta}\|$. Idempotence and mutual orthogonality follow from the group law and the orthogonality of the characters of T^2 . Commutation with T and its derivatives is (H2). For (32), let K_N

denote the two-dimensional Fejér kernel, a nonnegative approximate identity on T^2 , and set

$$\sigma_N u = \iint K_N(\alpha, \beta) \mathcal{R}_{\alpha\beta} u d\alpha d\beta, \quad (33)$$

a finite positive combination of the $\Pi_{m_1 m_2} u$ with weights $(1 - \frac{|m_1|}{N+1})(1 - \frac{|m_2|}{N+1})$. Strong continuity and the approximate-identity property give $\sigma_N u \rightarrow u$ in norm. If every $\Pi_{m_1 m_2} u$ vanishes then $\sigma_N u = 0$ for all N , hence $u = 0$. $\square$

Remark 2. We deliberately do *not* assert that the Fourier series of u converges, nor that X is an unconditional direct sum of isotypic subspaces; in a general Banach representation neither is automatic, and neither is needed. Only boundedness, commutation and (32) enter the argument, and all three follow from (H1)–(H2) by Fejér summation. Hypotheses (H1)–(H2) hold in the standard realizations of the quasinormal problem — weighted spaces adapted to (24), and the hyperboloidal realization [10] — because (30) relabels two *compact* coordinates, acts by isometries of (3), touches neither r nor θ , and leaves (21) and (24) invariant, the horizon exponent depending only on the sector's own labels. The decomposition is over (m_1, m_2) *only* and does not use separation of variables in θ ; see Sec. IV D.

B. Jordan chains and the order of the resolvent pole

Definition 3. A tuple $(v_0, \dots, v_{q-1})$ with $v_0 \neq 0$ is a Jordan chain of length q for the holomorphic pencil T at ω_* if

$$\sum_{i=0}^j \frac{1}{i!} T^{(i)}(\omega_*) v_{j-i} = 0, \quad j = 0, \dots, q-1. \quad (34)$$

The eigenvalue is *semisimple* if every chain has length one and *defective* otherwise; an EP of order q is a parameter value at which the maximal chain length is $q \geq 2$.

By the Keldysh expansion (Appendix C), the maximal chain length at ω_* equals the order of the pole of $T(\omega)^{-1}$ there, hence by (1) the degree of the secular polynomial in the ringdown. The following lemma therefore has direct physical content.

Lemma 4. *Under Proposition 1, every Jordan chain of T at ω_* projects componentwise onto Jordan chains of the sector restrictions $T_{m_1 m_2}$, at least one of which is nontrivial and of the same length; conversely a chain in one sector extends by zero. Hence*

$$\max_{\text{chains of } T} q = \max_{(m_1, m_2)} \max_{\text{chains of } T_{m_1 m_2}} q. \quad (35)$$

Proof. Apply the bounded projector $\Pi_{m_1 m_2}$ to the j -th equation of (34). Commutation with every $T^{(i)}$ gives

the identical relation for the projected components, so $(\Pi_{m_1 m_2} v_0, \dots, \Pi_{m_1 m_2} v_{q-1})$ is a chain in that sector. By (32), $v_0 \neq 0$ forces $\Pi_{m_1 m_2} v_0 \neq 0$ for at least one sector, whose chain then has length q . The converse is immediate, and (35) follows. $\square$

Theorem 5 (Cross-sector no-go). *A coincidence of quasinormal frequencies belonging to distinct azimuthal sectors cannot by itself create defectiveness. Any Jordan chain of length greater than one at ω_* must already occur within a single fixed (m_1, m_2) sector. Equivalently, such a coincidence cannot raise the order of the pole of $T(\omega)^{-1}$, hence cannot produce a secular term in (1).*

Proof. Immediate from Lemma 4 and Appendix C. $\square$

Remark 6. The wording is chosen so as not to exclude the case in which two sectors share a frequency while one of them independently contains an exceptional point. There T is defective, but the defectiveness belongs to the single sector and owes nothing to the coincidence. The underlying algebra is standard [5]; what is specific here is that Proposition 1 holds for *all* a, b , so the conclusion is not confined to the symmetric locus.

C. The equal-spin multiplets

Proposition 7. *On the equal-spin surface the labels degenerate at fixed (ℓ, m) form*

$$\mathcal{M}_{\ell, m} = \{(m_1, m_2) : m_1 + m_2 = m, |m_1| + |m_2| \leq \ell, \quad (36)$$

$\ell - |m_1| - |m_2| \in 2\mathbb{Z}_{\geq 0}\}, \text{ with } |\mathcal{M}_{\ell, m}| = \ell + 1 \text{ independently of } m; \text{ summing over the } \ell + 1 \text{ admissible } m \text{ gives } (\ell + 1)^2.$

Proof. For $m \geq 0$ put $m_1 = m - m_2$. The value $|m_1| + |m_2| = m$ is attained for $m_2 \in [0, m]$, that is $m + 1$ pairs, while each value $m + 2j$ with $j \geq 1$ is attained by exactly two pairs, $m_2 = m + j$ and $m_2 = -j$. The constraint in (36) admits $j \leq (\ell - m)/2$, whence $|\mathcal{M}_{\ell, m}| = (m + 1) + 2 \cdot \frac{\ell - m}{2} = \ell + 1$, independent of m . The case $m < 0$ follows from the parity generator of Sec. V. The admissible m are $-\ell, -\ell + 2, \dots, \ell$, so the level total is $(\ell + 1)^2$, the dimension of the $SO(4)$ scalar harmonic level consistent with (18). $\square$

By (19) the members of $\mathcal{M}_{\ell, m}$ share the same radial problem at $\delta = 0$ and are exactly degenerate there; they carry distinct (m_1, m_2) and therefore lie in distinct sectors. By Theorem 5 that coincidence cannot hybridize into a shared defective block at any δ . It is essential to state what this does *not* say: the frequencies are free to separate, and they do. Figure 1 shows the sectors (1, 1) and (2, 0), which share $m = 2$, meeting at $\delta = 0$ and separating for $\delta \neq 0$, the splitting being $\mathcal{O}(s\delta)$ as (17) requires. The protection is against hybridization, not against splitting.

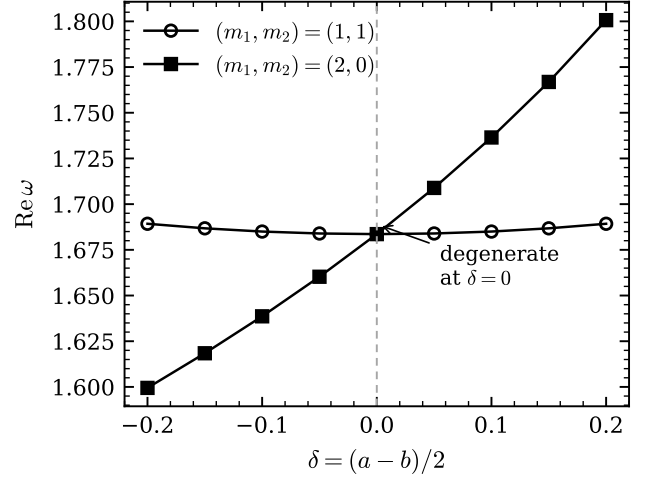


FIG. 1. Sectors (1, 1) and (2, 0) share $m = m_1 + m_2 = 2$ and are exactly degenerate on the equal-spin surface, separating for $\delta \neq 0$ ($\ell = 2$, $N = 0$, $s = 0.20$, $\mu = 0.6$). By Theorem 5 the coincidence at $\delta = 0$ cannot by itself be defective; the branches nevertheless split freely as δ varies.

D. What survives

Within a single sector, distinct angular branches ℓ are *not* separate summands: $\Lambda_j(\omega)$ depends on ω , so they are branches of one analytic family and Lemma 4 does not apply to them. This is why Proposition 1 was established without invoking separation of variables: a decomposition taken over (m_1, m_2, ℓ) would have proved too much, and would have been false. Collisions between distinct (ℓ, N) branches inside a fixed sector are the surviving channel, and the rest of the paper addresses them.

V. RESIDUAL SYMMETRY AND A CONDITIONAL TERMINATION LAW

Two exact transformations act on the pair (parameters, sector labels),

$$E : (a, b, m_1, m_2) \mapsto (b, a, m_2, m_1), \quad (37)$$

$$P : (a, b, m_1, m_2) \mapsto (-a, -b, -m_1, -m_2), \quad (38)$$

where E interchanges the labels of the two rotation planes and P is the orientation reversal $\phi \rightarrow -\phi$, $\psi \rightarrow -\psi$ composed with reversal of both rotation senses. Both leave (3) and (13)–(15) invariant for arbitrary a, b, μ , and

$$G = \{1, E, P, EP\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2, \quad (39)$$

acting in adapted coordinates as $E : (s, \delta) \mapsto (s, -\delta)$, $P : (s, \delta) \mapsto (-s, -\delta)$, $EP : (s, \delta) \mapsto (-s, \delta)$.

The essential observation is that *no element of G acts on the parameters alone*: each also moves the sector. A symmetry of the parameter space *at fixed sector* therefore exists only in sectors stabilized by the corresponding

element, as collected in Table II. In particular the conventional assertion that exceptional sets come in quadruples under independent reflections $s \rightarrow -s$ and $\delta \rightarrow -\delta$ holds only in the sector $(0, 0)$.

In a diagonal sector every branch is even in δ , hence — granted analyticity in the parameters — an analytic function of

$$t \equiv \delta^2. \quad (40)$$

Let $\omega_{\pm}$ denote two interacting branches and define the discriminant

$$D \equiv (\omega_+ - \omega_-)^2. \quad (41)$$

Then D depends on δ only through t , and an EP2 is $D = 0$: two real conditions, the familiar codimension two [6], defining a curve in (s, t, μ) . The physical parameter space, however, is the double cover $\delta = \pm\sqrt{t}$, which exists only for $t \geq 0$.

Proposition 8 (Mirror pairing and conditional termination). *Assume analyticity in the parameters and let Γ be a component of $\{D = 0\}$ in (s, t, μ) . The physical locus is invariant under E , so exceptional branches occur in mirror pairs. If in addition Γ meets $t = 0$ transversally, passing into $t < 0$, then parametrizing Γ by arclength τ with $t \approx c\tau$, $c \neq 0$,*

$$\delta = \pm\sqrt{c\tau}, \quad (42)$$

so the two mirror branches meet at $\delta = 0$ and terminate, approaching the equal-spin surface quadratically rather than crossing it.

The termination is conditional on transversality: a tangential crossing, or a component remaining in $t > 0$, is not excluded. This is the mechanism by which the equal-spin symmetry constrains exceptional topology here — not by protecting a degeneracy, that channel being closed by Theorem 5, but by constraining the *unfolding*. It is the analogue of the symmetry-constrained exceptional structures catalogued in non-Hermitian lattice models [28, 29], differing in that symmetry there lowers the codimension of higher-order points, whereas here it acts on the real-ity structure of the unfolding parameter. The prediction was recorded before the numerical search and remains untested, no exceptional locus having been found to exhibit it.

sector	stabilizer	induced symmetry
$m_1 = m_2$	E	$\omega(s, \delta) = \omega(s, -\delta)$
$m_1 = -m_2$	EP	$\omega(s, \delta) = \omega(-s, \delta)$
$(0, 0)$	G	even in δ and in s
generic	$\{1\}$	none

TABLE II. Stabilizers and the induced parameter symmetries. All three nontrivial cases are verified numerically below 10^{-12} , with a negative control confirming that a generic sector is *not* even in δ (Supplemental Material).

VI. LOCAL SPECTRAL THEORY AND ADMISSIBLE DIAGNOSTICS

A. Normal form, Newton polygon and Puiseux expansion

Two interacting branches are described locally by a 2×2 non-Hermitian generator $H(\mathbf{p})$. Its individual entries are not observable, since any similarity transformation alters them, but the two symmetric functions are:

$$T = \omega_+ + \omega_- = \text{tr } H, \quad D = (\text{tr } H)^2 - 4 \det H. \quad (43)$$

The eigenvalues are $\omega_{\pm} = \frac{1}{2}(T \pm \sqrt{D})$, so an EP2 is exactly $D = 0$. Writing the characteristic equation as $\mathcal{P}(\omega, t) = \omega^2 - T\omega + \det H = 0$ near a simple zero of D along a transversal direction t , the Newton polygon of $\mathcal{P}$ has a single segment of slope $1/2$, and the branches admit the Puiseux expansion

$$\omega_{\pm}(t) = \omega_* + c_1 t \pm c_{1/2} t^{1/2} + c_{3/2} t^{3/2} + \mathcal{O}(t^2), \quad (44)$$

with $c_{1/2} \neq 0$ [5, 44]. Three observable signatures follow, and we use all three:

1. the splitting obeys $|\omega_+ - \omega_-| \simeq 2|c_{1/2}|t^{1/2}$, against $\mathcal{O}(t)$ at an ordinary crossing and a nonzero constant at an avoided crossing;
2. continuing the pair once around a small loop encircling the EP interchanges the branches, and two circuits restore them, so the monodromy is the transposition of S_2 ;
3. the argument principle counts two zeros inside a contour that shrinks with the candidate.

For an EP of order q the Newton polygon has slope $1/q$, the splitting scales as $t^{1/q}$, and the monodromy is a q -cycle.

B. Two inadmissible diagnostics

Because the conclusion of Sec. IX is negative, the admissibility of each diagnostic must be established rather than assumed. We record two natural choices that are *not* admissible.

The derivative of the spectral condition. F in (29) is determined only up to a nonvanishing holomorphic factor: Leaver's continued fraction may be rescaled, or inverted at a different depth, without changing its zeros. Under $F \mapsto gF$ one has $\tilde{F}'(\omega_1) = g(\omega_1)F'(\omega_1)$ at a simple root, so $|dF/d\omega|$ has no invariant meaning and cannot bound the distance to an exceptional point.

The ratio of leading Taylor coefficients. Expanding $F = \sum_k a_k(\omega - \omega_1)^k$ about a simple root, $|a_1/a_2|$ approximates the distance to a nearby second root. It fails as a bound for two independent reasons. First, it is

only *asymptotically* insensitive to normalization: writing $g = g_0 + g_1(\omega - \omega_1) + \dots$,

$$\frac{\tilde{a}_1}{\tilde{a}_2} = \frac{a_1/a_2}{1 + (g_1/g_0)(a_1/a_2)}, \quad (45)$$

so the distortion is $\mathcal{O}(|g'/g|d)$ with d the separation. This is measurable: for $g = Ae^{k\omega}$ the observed ratio follows $|1 + kd|^{-1}$ to five decimals, from 1.0003 at $d = 10^{-4}$ to 1.43 at $d = 0.1$ [Fig. 2(a)]. Second, and decisively, $|a_1/a_2|$ measures the distance to the nearest zero *or pole*. Leaver's continued fraction is meromorphic, and contour integration about our tightest candidate returns $\mathcal{N} - \mathcal{P} = 0$ at every radius out to 0.20: one zero and one pole enclosed. The small values returned are pole distances and carry no spectral meaning. We therefore *withdraw* this estimator as a bound and retain it only as a local indicator.

What survives is admissible because it is built from eigenvalue differences of independently verified modes: the pairwise separation between tracked branches, each confirmed by an ODE residual of order 10^{-12} and by cross-solver agreement; and the eigenvalue condition number

$$\kappa = \frac{\|u\| \|v\|}{|v^\dagger T'(\omega_*)u|}, \quad (46)$$

computed from Solver C's collocation matrix, which is free of the continued fraction's poles. By the Keldysh expansion, $\kappa^{-1} \propto |v^\dagger T'u|$ is precisely the residue normalization of the simple pole, so $\kappa \rightarrow \infty$ is the analytic signature of an approaching EP. We note also that the discriminant (41) is *not* independent evidence, being the square of the separation.

C. The verification gate

A detector that has never been shown to fire cannot support a negative result. All routines were calibrated on problems whose exceptional structure is known in closed form — $\lambda^2 - t$ (EP2), $\lambda^3 - t$ (EP3), $\lambda^2 - (t^2 + g^2)$ (avoided crossing), Jordan blocks, and block-diagonal coincidences — and, equally importantly, on *negative controls* in which the detector must not fire [Fig. 2(b)]. Roots are recovered from contour moments by Newton's identities, requiring no starting guess [45]. A candidate is accepted only if it satisfies every applicable condition of Table III.

	condition	ten tightest
G1	separation $\rightarrow 0$ under refinement	fail
G2	argument-principle count = 2	pass
G3	both roots found without a guess	pass
G4	monodromy interchanges the branches	fail
G5	two circuits restore the assignment	—
G6	splitting $\sim t^{1/2}$, preferred over t^1	fail
G7	Jordan chain solvable, geom. < alg.	—
G8	stable under resolution/depth/precision	—
G9	confirmed by a recurrence-free solver	—
G10	inside the validated solver domain	pass

TABLE III. The EP2 gate. G2 and G3 are *preconditions*: they establish only that two distinct roots lie near one another, which is what makes a point a candidate. The decisive evidence is G1, G4 and G6, which fail uniformly. For an EP3 one additionally requires three coalescing branches, count 3, $a_1 = a_2 = 0$ with $a_3 \neq 0$, cube-root splitting, and a three-cycle monodromy.

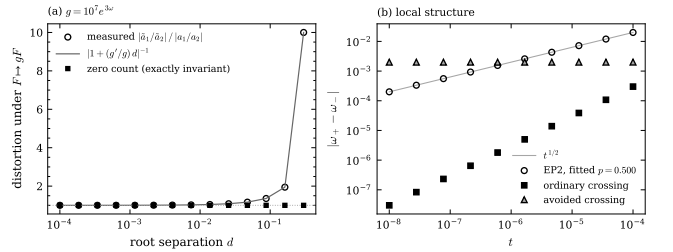


FIG. 2. (a) The coefficient ratio is only asymptotically insensitive to the normalization of the spectral condition: its distortion under $F \mapsto 10^7 e^{3\omega} F$ follows $|1 + (g'/g)d|^{-1}$ exactly, reaching 43% at $d = 0.1$, whereas the enclosed zero count is unaffected. (b) The Puiseux exponent separates an EP2 ($t^{1/2}$) from an ordinary crossing (t^1) and from an avoided crossing, which saturates at a constant.

VII. NUMERICAL FORMULATION

A. Leaver's method

Factoring the boundary behaviour of Sec. III C out of (14),

$$R = (z - z_+)^{-i\sigma_+} (z - z_-)^{\xi} e^{i\Omega r} r^{-3/2} \sum_{n \geq 0} a_n \left(\frac{z - z_+}{z - z_-} \right)^n, \quad (47)$$

and clearing denominators yields a linear recurrence with polynomial coefficients [46, 47],

$$\sum_{j=0}^p c_j(n) a_{n-j} = 0, \quad (48)$$

of width $p + 1 = 13$ for generic two-spin parameters, collapsing to 9 in the static case. Gaussian elimination reduces (48) to three terms, $\alpha_n a_{n+1} + \beta_n a_n + \gamma_n a_{n-1} = 0$, and the quasinormal condition — that the *minimal* solu-

tion be selected — becomes the vanishing of a continued fraction,

$$\beta_N - \frac{\alpha_{N-1}\gamma_N}{\beta_{N-1}-} \dots = \frac{\alpha_N\gamma_{N+1}}{\beta_{N+1}-} \frac{\alpha_{N+1}\gamma_{N+2}}{\beta_{N+2}-} \dots \quad (49)$$

We stress a point that cost us: the inversion index N changes the conditioning of (49) and, at finite depth, its zero set. At a tracked $N = 2$ root the $N = 2$ inversion gives $|F| = 1.9 \times 10^{-13}$ while inversions 0, 1, 3 give 7×10^{-2} , 1×10^{-1} and 1.1, values unchanged from depth 200 to depth 3200. Every quantity extracted from the wrong inversion is void, and a residual gate now enforces the correct choice.

B. Large- n asymptotics

The convergence of (49) is governed by the large- n behaviour of $R_n = a_n/a_{n-1}$. Writing the characteristic equation of (48) as $A(1/R) = 0$, the root $x = 1$ has multiplicity at least five in A and exactly four in the neighbouring coefficient polynomials, so the integer-power ansatz degenerates and half-integer powers survive:

$$R_n = 1 + u_1 n^{-1/2} + u_2 n^{-1} + \mathcal{O}(n^{-3/2}), \quad u_1 = -\sqrt{-2c}, \quad (50)$$

with $c = i\Omega(r_+ - r_-)$ and the sign fixed by minimality. Including the leading term reduces the depth required for a given accuracy by a factor ≈ 1.35 , worth 1.4 digits at depth 200. Higher coefficients are unavailable: the Gaussian reduction degrades beyond $n \sim 500$ –800, where the backward ratio recursion collapses onto a spurious, problem-independent $R_n - 1 = 2/n$. See Appendix B.

C. Exterior complex scaling

An independent formulation rotates the radial contour beyond a matching radius, $r = r_m + \varsigma e^{i\vartheta}$, $\varsigma \in [0, L]$, and discretizes (14) by Chebyshev collocation. The outgoing solution (24) decays along the contour provided

$$\tan \vartheta > -\frac{\text{Im } \Omega}{\text{Re } \Omega}, \quad (51)$$

defining a minimum admissible angle $\vartheta_{\min}$. Under this transformation a genuine resonance becomes an eigenvalue of the scaled operator and is independent of ϑ , whereas the rotated continuum sweeps with ϑ ; the construction thus separates resonances from artifacts without any root finder. Condition (51) is decisive in Sec. X.

D. Solvers, domains and benchmarks

Four solvers are used, in two families sharing no discretization (Table IV). Any claim requires agreement between the families; two implementations wrapping one

discretization are not independent evidence. Related matrix and spectral methods are reviewed in [48–53], with open implementations for Kerr in [54].

The five-dimensional Schwarzschild–Tangherlini fundamental scalar mode is reproduced to all twelve published digits,

$$\omega = 0.533835574268 - 0.383375368513i \quad (r_+ = 1), \quad (52)$$

with absolute error 5.0×10^{-13} against Ref. [55] and 2.2×10^{-12} between Solvers A and B. The exchange symmetry $(a, m_1) \leftrightarrow (b, m_2)$ holds numerically to 4.2×10^{-15} , and three solvers agree below 10^{-5} on ten benchmark points. The atlas domain extends to $r_- = 0.2287$, beyond the previously validated $r_- \leq 0.19$; Solvers A and C agree to at most 2.06×10^{-9} on nine spanning points including the domain corners, which is the evidence supporting that extension.

VIII. THE BRANCH ATLAS

Branches are identified by continuation history, never by sorting frequencies: at an avoided crossing two branches exchange order in $\text{Re } \omega$ while remaining distinct modes. Each step is seeded by quadratic extrapolation in an arclength parameter and accepted only if the predictor miss $\varepsilon_{k+1} = |\omega_{k+1} - \omega_{k+1}^{\text{pred}}|$ satisfies

$$\varepsilon_{k+1} \leq \max\left(4 \text{ med}\{\varepsilon_1, \dots, \varepsilon_k\}, 0.02 \max(|\omega_{k+1}|, 1)\right). \quad (53)$$

The first argument is essential: the predictor miss is a local truncation error and follows a smooth trend along a genuine branch, so a jump appears as a sudden multiple of the running median. A tolerance proportional to $|\omega|$ alone is a *constant*, and at $|\omega| \approx 3.7$ it admitted a jump in which the trend of $\text{Im } \omega$ reversed; the contaminated point became, until detected, the tightest reported candidate.

The atlas comprises 50 558 points over the seven independent sectors (0, 0), (1, 1), (−1, −1), (1, 0), (2, 0), (2, 1), (2, 2), with $\ell \in \{\ell_{\min}, \ell_{\min} + 2, \ell_{\min} + 4\}$, $\ell_{\min} = |m_1| + |m_2|$, and $N = 0, \dots, 3$, over

$$s \leq 0.42, \quad |\delta| \leq 0.20, \quad \mu \leq 1.8, \quad r_- \leq 0.2287, \quad \mathcal{E} \geq 0.294. \quad (54)$$

The remaining sectors (0, 1), (0, 2), (1, 2) are images of (1, 0), (2, 0), (2, 1) under E and are spot-checked rather

formulation	rec.-free	domain
A Leaver continued fraction	no	$r_- \leq 0.2287$
B raw recurrence, Hill, Wynn	no	as A
C exterior complex scaling	yes	as A, $\vartheta_{\min} < 90^\circ$
D multidomain near-horizon	yes	ordinary modes

TABLE IV. Solver inventory. Solver C’s independence is enforced structurally: a test inspects the module and fails if any recurrence routine is referenced.

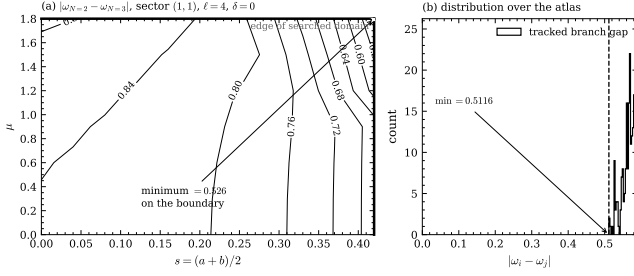


FIG. 3. The bounded exclusion. (a) Level curves of the separation between the $N = 2$ and $N = 3$ branches at $\ell = 4$ in sector $(1, 1)$, on the equal-spin surface; the star marks the global minimum, which lies on the boundary of the searched domain (heavy axes). (b) Distribution of the tracked separation over the atlas. No claim is made outside the domain (54).

than recomputed. The maximum continued-fraction residual is 3.6×10^{-9} ; 118 predictor-miss and 33 non-convergence events are recorded rather than filled in, and four branch-label collapses are excluded and counted.

Two predictions of Secs. IV–V are recovered from independently computed frequencies as an end-to-end check, having been imposed nowhere:

$$|\omega_i - \omega_j|_{+\delta} - |\omega_i - \omega_j|_{-\delta} \lesssim 10^{-12}, \quad (55)$$

$$|\text{gap}_{(1,1)} - \text{gap}_{(2,0)}|_{\delta=0} = 9 \times 10^{-12}. \quad (56)$$

IX. THE BOUNDED EXCLUSION

Within the domain (54),

$$\min_{\text{atlas}} |\omega_i - \omega_j| = 0.5116, \quad \kappa_{\text{max}} = 8.73 \times 10^3, \quad (57)$$

the latter against $\kappa = 18.3$ at an ordinary control mode. An EP2 requires the separation to vanish and, by (46) and the Keldysh expansion, κ to diverge; neither occurs. The minimum separation is ≈ 0.51 where $|\omega| \approx 2-4$, a relative separation of order 15%, and it is attained on the boundary of (54), at $s = 0.42$, $\delta = -0.20$, $\mu = 1.8$ (Fig. 3).

The ten tightest candidates were put through the gate of Table III. All ten are rejected with a common signature: two distinct roots are present, but there is no coalescence, no monodromy interchange, and no $t^{1/2}$ splitting. The classification is therefore *avoided crossing* — the behaviour familiar from eigenvalue repulsion in the Kerr–Newman family [14, 16], here without the accompanying exceptional points. Consistently, the elevated but finite $\kappa = 8.73 \times 10^3$ is what one expects at a genuine avoided crossing: by (46) it is a factor 476 above the control value, whereas an approaching EP2 would show $\kappa \sim d^{-1}$ with $d \rightarrow 0$.

X. THE QUASIRESONANT LIMIT

As μ grows, branches migrate toward the threshold $\omega^2 = \mu^2$ and become arbitrarily long lived, the five-dimensional analogue of the quasis resonances of massive fields on rotating backgrounds [27, 35, 36]. Tracking eleven branches across six sectors with an adaptive step, the damping falls through 8.6 orders of magnitude along a single continuous branch, from 0.357 at $\mu = 0$ to 8.65×10^{-10} at $\mu = 2.968$, with $\text{Re } \omega \rightarrow \mu$. This is a branch point in μ , not a coalescence: the neighbouring overtone remains at $-\text{Im } \omega \sim 0.1$ throughout, so the branches separate rather than merge. Every walk terminates where double precision can no longer resolve the mode, so the reported minima are lower bounds on how far a branch was followed, not infima.

Because $\text{Re } \omega \rightarrow \mu$ from above while $\omega_c \leq 0.51$ in these sectors, the quasis resonant branches move *away* from the superradiant condition (27) as μ grows. The long-lived modes we track are therefore not incipient superradiant instabilities, and the mechanism that would produce a genuine instability — a quasi-bound state with $\text{Re } \omega < \omega_c$, requiring μ small and ω_c large — lies outside (54).

The same regime settles a question about solver behaviour that proves to be physics rather than numerics. By (51), as $\text{Re } \Omega \rightarrow 0$ the outgoing wave ceases to oscillate and becomes a pure growing exponential, so

$$\vartheta_{\min} \rightarrow 90^\circ \quad \text{as} \quad \text{Re } \Omega \rightarrow 0, \quad (58)$$

and *no admissible contour exists*. We measure $\vartheta_{\min} = 89.92^\circ$ and 89.81° at the two points where the recurrence-free solvers fail, against 4.4° at an ordinary near-extremal point where they succeed. Equation (58) is a limit of the *formulation*, not a convergence failure: no increase of resolution, precision or domain decomposition removes it, which is why a multidomain refinement of the near-horizon region — which resolves $r_+ - r_-$ by construction — did not help. It follows that difficulty in this range is controlled by μ and *not* by proximity to extremality: ordinary modes continue smoothly to $\mathcal{E} = 0.0103$ with residuals $\sim 10^{-13}$ over thirty-seven consecutive steps.

XI. OBSERVATIONAL AND ASTROPHYSICAL CONTEXT

The system studied here is not itself an astrophysical object. Its relevance is methodological and structural, and we state the connection carefully rather than overstate it.

First, the observational stakes of exceptional structure lie in ringdown spectroscopy. Multimode fits to a post-merger signal presuppose a discrete, diagonalizable spectrum with well-separated overtones; by (1) an exceptional point replaces two exponentials by $(A + Bt)e^{-i\omega_* t}$, and near one the individual excitation factors diverge while the total response stays finite, so that fitted amplitudes

become ill-conditioned [9, 20, 22]. Any programme that tests the no-hair theorem by extracting several damped exponentials therefore has a stake in whether exceptional points occur at accessible parameters, and in whether symmetry generates them systematically.

Second, our negative result speaks to a specific mechanism. The natural conjecture — that an enhanced isometry produces degeneracies which unfold into exceptional points — is exactly what Theorem 5 forbids whenever the degenerate modes are distinguished by conserved azimuthal quantum numbers. The theorem is not special to five dimensions: it uses only that the background possesses two commuting azimuthal Killing fields that survive the symmetry breaking. In four-dimensional Kerr the corresponding statement, with the single Killing field ∂_ϕ , forbids exceptional points between modes of different m ; the exceptional points that are found in massive-scalar Kerr [17, 18] are, consistently, within a single m sector. Searches for symmetry-protected exceptional structure should therefore be directed at within-sector overtone collisions rather than at multiplet degeneracies.

Third, higher-dimensional rotating black holes are of interest as the setting where uniqueness fails and where ultraspinning and Gregory–Laflamme instabilities operate [39, 40]. Our domain (54) is far from the ultraspinning regime — it is bounded by $\mathcal{E} \geq 0.294$, whereas the relevant instabilities set in near extremality — so the modes studied here are stable, damped, and non-superradiant throughout, as Table I and Sec. IIID confirm.

XII. LIMITATIONS

The exclusion is bounded, and we state the boundaries plainly. The minimum lies on the edge of (54); a targeted probe beyond it finds the separation falling to 0.472 at $s = 0.45$ but rising again at $\mu = 3.0$ and 3.6, so the trend does not extrapolate to zero — yet no claim is made for $s > 0.45$ or $\mu > 1.8$. Overtones $N > 3$ and higher ℓ were not searched, and highly damped overtones are known to be delicate [56, 57]. The superradiant regime (27) was not searched, and there the horizon boundary condition changes character. The quasis resonant regime was not searched, since by (58) no recurrence-free solver is applicable and the evidentiary standard of Table III cannot be met; this is the one region of the studied sectors in which an exceptional point could still be hiding.

Certification was attempted and is limited in scope [58–61]: we obtain a Krawczyk certificate for finite-dimensional systems and an Arb ball enclosure of the angular eigenvalue of the truncated Jacobi matrix, validated by recovering $\ell(\ell+2)$ at $c^2 = 0$. The radial problem is *not* certified, its recurrence coefficients being extracted by a discrete Fourier transform for which we have no rigorous aliasing bound, so the truncation term in

$$E_{\text{total}} \leq E_{\text{disc}} + E_{\text{trunc}} + E_{\text{arith}} + E_{\text{root}} \quad (59)$$

cannot be bounded. No truncated-recurrence or continuum certificate is claimed.

Finally, the numerical result is an exclusion over a declared domain, not a theorem, and must not be conflated with Theorem 5, which is exact.

XIII. DISCUSSION

We exclude exceptional points within the declared numerical domain; we do not claim their absence from the system. Exceptional lines *do* occur in the de Sitter members of this family [25], and the present $\Lambda = 0$ result is a contrast between spacetimes rather than a contradiction: the de Sitter mechanism relies on the cosmological horizon and the Pöschl–Teller reduction, neither of which has an asymptotically flat analogue, and the accompanying eigenvalue repulsions [26] are likewise a de Sitter phenomenon.

Within the asymptotically flat doubly rotating case the structure is sharply organized. The most natural route to exceptional behaviour is closed exactly, by the same symmetry that generates the candidate degeneracies. The surviving route is numerically weak in the region searched: those branches repel with a separation never falling below half a unit. Since the parameter space has exactly the dimension at which codimension-two sets should be generic [6], their absence in this region is informative rather than expected.

Two methodological points may outlive the application. First, neither the derivative of a spectral condition nor the ratio of its leading Taylor coefficients is an admissible bound on the distance to an exceptional point: the former is not invariant under the renormalization freedom, and the latter measures the distance to the nearest zero *or pole* of a meromorphic condition. Second, a small computed separation between two *labels* is evidence about the labels before it is evidence about the physics.

The natural next step is a hyperboloidal formulation, the only method that remains applicable as $\text{Re } \Omega \rightarrow 0$ and therefore the only route into the quasis resonant regime [8, 10], together with an extension into the superradiant sector, where the boundary condition must be re-derived and where a genuine instability, rather than an exceptional point, is the object of interest.

ACKNOWLEDGMENTS

ACKNOWLEDGMENTS NOT ASSIGNED.

Appendix A: Separation of the massive Klein–Gordon equation

With the ansatz (12), the equation $\partial_\nu(\sqrt{|g|}g^{\nu\rho}\partial_\rho\Phi) = \mu^2\sqrt{|g|}\Phi$ requires the inverse metric. Rather than inverting the full 5×5 matrix, note that (3) is block di-

agonal: the pair (r, θ) is diagonal, and the remaining (t, ϕ, ψ) block is 3×3 . Inverting the small block and using $\sqrt{|g|} = r \Sigma \sin \theta \cos \theta$ gives the wave operator in closed form.

Substituting $u = \cos^2 \theta$ converts the result into a rational function of (r, u) , at which point exact cancellation is tractable. Writing the separation quotient as

$$Q(r, u) = \frac{1}{R} \mathcal{L}_r R + \frac{1}{S} \mathcal{L}_u S, \quad (\text{A1})$$

separability is the statement $\partial^2 Q / \partial r \partial u \equiv 0$, which we verify in exact rational arithmetic. Partial fractions in u then isolate the angular operator, giving (13); the remainder is (14). Two checks are applied: $\sqrt{|g|}$ must equal $r \Sigma \sin \theta \cos \theta$ identically, and at $a = b = \mu = 0$, $m_1 = m_2 = 0$ the radial bracket must reduce to $\omega^2 r^4 / (r^2 - M) - \Lambda$ with $\Lambda = \ell(\ell + 2)$.

Appendix B: Frobenius exponents and the asymptotic tail

Horizon. Near $z = z_+$, retaining the leading behaviour of $W^2 / (r^4 \Delta)$ in (14) gives the indicial equation (20). The factorization $W(z_+) = (z_+ + a^2)(z_+ + b^2)(\omega - \omega_c)$, which follows from (15) and (9), is what puts σ_+ in the form (21); the quasinormal condition selects $-\mathrm{i}\sigma_+$.

Infinity. At large r the equation becomes $R'' + \frac{3}{r} R' + (\Omega^2 + \mathcal{O}(r^{-2}))R = 0$. Because the potential is even in r there is no $\mathcal{O}(1/r)$ term, so the WKB exponent contains no logarithm and (24) follows, the outgoing branch being selected. The absence of a Coulomb phase distinguishes this problem from four-dimensional massive Kerr.

Tail. Setting $R_n = a_n / a_{n-1}$ in (48) and substituting

$$a_n \sim \exp\left(\lambda \sqrt{n} + \eta \log n + d_1 n^{-1/2} + d_2 n^{-1} + \dots\right) \quad (\text{B1})$$

shows that the characteristic polynomial $A(1/R)$ has $x = 1$ as a root of multiplicity at least five, while the neighbouring coefficient polynomials have multiplicity exactly four. The balance at order $n^{-3/2}$ reads $-u_1 A'(1) = 0$ with $A'(1) = 0$, so the integer-power series degenerates and half-integer powers survive, giving (50). Two independent numerical confirmations are available: the scaling test, in which $(R_n - 1)\sqrt{n}$ converges while $(R_n - 1)n$

diverges, and the depth signature $\exp(-k\sqrt{N})$ of the truncation error.

Appendix C: Keldysh expansion and the ringdown pole order

Let $T(\omega)$ be holomorphic and Fredholm of index zero near ω_* , with a single eigenvalue there. If ω_* is *simple*, with right and left null vectors u, v normalized by $v^\dagger T'(\omega_*)u = 1$, then

$$T(\omega)^{-1} = \frac{u v^\dagger}{\omega - \omega_*} + \mathcal{O}(1), \quad (\text{C1})$$

a first-order pole whose residue is the rank-one spectral projector. More generally, if the canonical system of root functions at ω_* has maximal chain length q , the Keldysh theorem gives

$$T(\omega)^{-1} = \sum_{k=1}^q \frac{\mathcal{S}_k}{(\omega - \omega_*)^k} + \mathcal{O}(1), \quad (\text{C2})$$

with $\mathcal{S}_q \neq 0$ built from the chain vectors. The order of the pole is therefore exactly the maximal Jordan-chain length.

The retarded Green function of the perturbation problem is $G(\omega) = T(\omega)^{-1}$ acting on the source, and the time-domain response is its inverse Laplace transform. A pole of order k at ω_* contributes $t^{k-1} e^{-\mathrm{i}\omega_* t} / (k-1)!$, giving (1). Combining with Lemma 4: since the maximal chain length of T equals the maximum over sectors, and a cross-sector coincidence contributes no chain longer than those already present in the individual sectors, such a coincidence cannot raise the pole order and cannot produce a secular term. This is the precise sense in which Theorem 5 has observational content.

Equation (C1) also explains the role of (46): the normalization $v^\dagger T'(\omega_*)u$ is the residue denominator, so $\kappa \propto |v^\dagger T' u|^{-1}$ diverges precisely when the residue of the simple pole blows up, i.e. as two simple poles merge into a double one.

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